STRUCTURAL ANALYSIS / TECHNICAL REPORT

Manual load combination

Simply supported beam with permanent and variable uniform loads combined manually.

Analysis id: ULS1

Source: examples/combinations/load-combination.json

Generated: 2026-10-19 00:03 +02:00

01 Model overview

2	1	2	2	2	2
Nodes	Members	Supports	Load cases	Combinations	Loads

GEOMETRY & DEFORMATION

Grey: original geometry / Teal: auto-amplified deformation

02 Executive summary

Nodal max |Uy| [m]	Max |V| [kN]	Max |M| [kNm]
0.000000	48.750000	60.937500

Metric	Entity / member	Pos [0-1]	x [m]	Value	Unit
Max |Ux|	A			0.000000	m
Max |Uy|	A			0.000000	m
Max |Rz|	A			0.024182	rad
Max |Fx|	SA			0.000000	kN
Max |Fy|	SA			48.750000	kN
Max |Mz|	SA			0.000000	kNm
Max |N|	M1	0.000000	0.000000	0.000000	kN
Max |V|	M1	0.000000	0.000000	48.750000	kN
Max |M|	M1	0.500000	2.500000	60.937500	kNm

03 Assumptions and limitations

First-order linear elastic 2D analysis. Units: m, rad, kN and kNm. Summary values are absolute magnitudes; detail tables retain their signs.

This is a technical computation report. It does not replace engineering judgement or project-specific normative checks.

04 Nodal displacements

Node	Ux [m]	Uy [m]	Rz [rad]
A	0.000000	0.000000	-0.024182
B	0.000000	0.000000	0.024182

05 Support reactions

Support	Node	Fx [kN]	Fy [kN]	Mz [kNm]
SA	A	0.000000	48.750000	0.000000
SB	B	0.000000	48.750000	0.000000

06 Member end forces

Member	End	N [kN]	V [kN]	M [kNm]
M1	Start	0.000000	48.750000	0.000000
M1	End	0.000000	48.750000	0.000000

07 Internal-force samples

Member	Pos [0-1]	x [m]	N [kN]	V [kN]	M [kNm]
M1	0.000000	0.000000	-0.000000	48.750000	0.000000
M1	0.050000	0.250000	-0.000000	43.875000	11.578125
M1	0.100000	0.500000	-0.000000	39.000000	21.937500
M1	0.150000	0.750000	-0.000000	34.125000	31.078125
M1	0.200000	1.000000	-0.000000	29.250000	39.000000
M1	0.250000	1.250000	-0.000000	24.375000	45.703125
M1	0.300000	1.500000	-0.000000	19.500000	51.187500
M1	0.350000	1.750000	-0.000000	14.625000	55.453125
M1	0.400000	2.000000	-0.000000	9.750000	58.500000
M1	0.450000	2.250000	-0.000000	4.875000	60.328125
M1	0.500000	2.500000	-0.000000	0.000000	60.937500
M1	0.550000	2.750000	-0.000000	-4.875000	60.328125
M1	0.600000	3.000000	-0.000000	-9.750000	58.500000

Internal-force samples / continued

Member	Pos [0-1]	x [m]	N [kN]	V [kN]	M [kNm]
M1	0.650000	3.250000	-0.000000	-14.625000	55.453125
M1	0.700000	3.500000	-0.000000	-19.500000	51.187500
M1	0.750000	3.750000	-0.000000	-24.375000	45.703125
M1	0.800000	4.000000	-0.000000	-29.250000	39.000000
M1	0.850000	4.250000	-0.000000	-34.125000	31.078125
M1	0.900000	4.500000	-0.000000	-39.000000	21.937500
M1	0.950000	4.750000	-0.000000	-43.875000	11.578125
M1	1.000000	5.000000	-0.000000	-48.750000	0.000000

08 Displacement samples

Member	Pos [0-1]	x [m]	GUx [m]	GUy [m]
M1	0.000000	0.000000	0.000000	0.000000
M1	0.050000	0.250000	0.000000	-0.005743
M1	0.100000	0.500000	0.000000	-0.010882
M1	0.150000	0.750000	0.000000	-0.015416
M1	0.200000	1.000000	0.000000	-0.019345
M1	0.250000	1.250000	0.000000	-0.022670
M1	0.300000	1.500000	0.000000	-0.025391
M1	0.350000	1.750000	0.000000	-0.027507
M1	0.400000	2.000000	0.000000	-0.029018
M1	0.450000	2.250000	0.000000	-0.029925
M1	0.500000	2.500000	0.000000	-0.030227
M1	0.550000	2.750000	0.000000	-0.029925
M1	0.600000	3.000000	0.000000	-0.029018
M1	0.650000	3.250000	0.000000	-0.027507
M1	0.700000	3.500000	0.000000	-0.025391
M1	0.750000	3.750000	0.000000	-0.022670
M1	0.800000	4.000000	0.000000	-0.019345
M1	0.850000	4.250000	0.000000	-0.015416
M1	0.900000	4.500000	0.000000	-0.010882
M1	0.950000	4.750000	0.000000	-0.005743
M1	1.000000	5.000000	0.000000	0.000000